NeuroVision AI - Lean Canvas

Explainable Brain MRI Tumor Classification: Glioma, Meningioma, Pituitary & Normal Scans

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1. PROBLEM

- Severe shortage of certified neuro-radiologists in rural and tier-2/3 hospital centers.
- Long turnaround times (hours to days) for routine and emergency MRI brain scan triage.
- Diagnostic fatigue and high inter-observer variability when distinguishing subtle neoplastic margins.
- Existing AI tools act as opaque "black boxes" outputting raw class labels without anatomical reasoning.

EXISTING ALTERNATIVES

- Manual review by on-call hospital radiologists.
- Outsourced tele-radiology networks (expensive, bandwidth-dependent).
- Proprietary CAD systems (high licensing costs, closed-source).
- Heuristic thresholding or unexplainable legacy CNN models.

2. SOLUTION

- Automated 4-class Brain MRI classification: **Glioma**, **Meningioma**, **Pituitary**, and **Normal** tissue.
- **Explainable AI (Grad-CAM)**: Generates gradient-weighted visual saliency overlays highlighting tumor boundaries.
- **Automated PDF Radiology Summaries**: Instant diagnostic reports with dual MRI/ heatmap visuals and clinical guidance.
- **Zero-Footprint Web UI**: Browser-based tool with real-time radar scan animation and scan history registry.

8. KEY METRICS

- **Overall Test Accuracy**: 96.38% on held-out test cohort ($n = 1,600$).
- **Macro AUC**: 0.9880; per-class AUC 0.982 (Glioma), 0.984 (Meningioma), 0.995 (Pituitary), 0.991 (Normal).
- **Macro F1-Score**: 0.9632 (Weighted F1: 0.9639).
- **Inference Latency**: < 1.2s end-to-end (24ms model inference).
- **False Positive Rate**: Only 2.7% on healthy brain scans.

3. UNIQUE VALUE PROPOSITION

- Upload any axial brain MRI and receive an accurate 4-class differential diagnosis with visual anatomical proof in under 2 seconds.
- Free, open-source, built for clinical education and triage assistance with transparent limitation disclosure.

HIGH-LEVEL CONCEPT

- **"Spell-check for Brain MRIs"**: An instantaneous, explainable second opinion before surgical or therapeutic intervention.
- Automated neuro-radiological pre-triage assistant.

4. UNFAIR ADVANTAGE

- Targeted 2-stage transfer learning: **EfficientNetB0** optimized with domain-adapted fine-tuning on brain MRI morphology.
- Direct activation inspection from layer `top_conv` (1280 channels) with zero approximation.
- Integrated ReportLab vector PDF engine and SQLite EMR registry.
- Full reproducibility: zero data leakage, balanced test benchmark.

5. CHANNELS

- Cloud-hosted web application accessible on desktop, tablet, and mobile.
- Teaching hospitals, medical colleges, and residency training workshops.
- AICTE and BharatCares academic & healthcare partner networks.
- GitHub open-source repository and developer community.
- Future: PACS / DICOM viewer plugin and REST API microservice.

6. CUSTOMER SEGMENTS

- Radiologists and neuro-oncologists seeking rapid pre-triage second opinions.
- Medical students and radiology residents training on tumor morphology.
- Tier-2 & rural diagnostic clinics lacking 24/7 on-site neuro-radiology specialists.
- Tele-radiology diagnostic networks.

EARLY ADOPTERS

- Medical college teaching hospitals and radiology residency programs.
- Rural health centers needing rapid tele-triage for acute transfers.
- Academic biomedical imaging researchers.

7. COST STRUCTURE

- Cloud hosting and GPU compute for model training and inference.
- Data curation, annotation validation by certified radiologists, and dataset storage.
- Model retraining, drift monitoring, and multi-center clinical validation.
- Developer and open-source maintenance time (currently student-built).

9. REVENUE STREAMS (HYPOTHESES TO VALIDATE)

- **Free Public & Academic Tool**: Zero-cost educational tier supporting UN SDG 3 & SDG 9.
- **B2B Institutional Licensing**: Subscription-based decision-support integration for private diagnostic chains.
- **Tiered REST API**: Per-scan micro-billing for tele-radiology platforms requiring triage feeds.
- **CSR & Healthcare Grants**: BharatCares, IBM SkillsBuild, and digital healthcare equity grants.

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